PRACTICAL 3

Aim :

Integrate a public REST API (e.g., GitHub API) into the portfolio application to fetch and display a list of repositories dynamically. Use the Projects page built in Practical 2 as the integration point. Implement a loading spinner component shown while the request is in progress. Implement an error message component shown if the API call fails. Render at least the repository name and URL for each item returned.

Code :



```css
.header {
  padding: 2rem 1.5rem;
  color: white;
  text-align: center;
}

:root {
  --bg: #ffffff;
  --text: #111827;
  --text-h: #111827;
  --muted: #6b7280;
  --nav-bg: #f3f4f6;
  --nav-text: #111827;
  --toggle-bg: #ffffff;
  --toggle-text: #111827;
  --toggle-border: #d1d5db;
}

[data-theme='dark'] {
  --bg: #0f172a;
  --text: #e6eef8;
  --text-h: #e6eef8;
  --muted: #94a3b8;
  --nav-bg: #0b1220;
  --nav-text: #e6eef8;
  --toggle-bg: #111827;
  --toggle-text: #e6eef8;
  --toggle-border: #374151;
}

.app-root {
  background: var(--bg);
  color: var(--text);
  min-height: 100vh;
}

main {
  max-width: 720px;
  margin: 0 auto;
  padding: 1.5rem;
}

.about,
.skills {
  margin-bottom: 2rem;
}

.skills ul {
  list-style: none;
  padding: 0;
  display: flex;
  flex-wrap: wrap;
  gap: 0.5rem;
}

.skills li {
  background: #eef2ff;
  color: #2563eb;
  padding: 0.35rem 0.75rem;
  border-radius: 999px;
  font-size: 0.9rem;
}

.footer {
  text-align: center;
  padding: 1rem;
  font-size: 0.85rem;
  color: #6b7280;
  border-top: 1px solid #e5e7eb;
}
```

AWDF-LAB - Projects.jsx

```jsx
import { useEffect, useMemo, useState } from "react";

function Projects() {
  const username = "utsavshah-in";

  const [repos, setRepos] = useState([]);
  const [loading, setLoading] = useState(true);
  const [error, setError] = useState(null);
  const [search, setSearch] = useState("");

  const fetchRepos = async () => {
    try {
      setLoading(true);
      setError(null);

      const response = await fetch(
        `https://api.github.com/users/${username}/repos?sort=updated&per_page=100`
      );

      if (!response.ok) {
        throw new Error(`GitHub API error: ${response.status}`);
      }

      const data = await response.json();
      setRepos(data);
    } catch (err) {
      setError(err.message || "Failed to fetch repositories");
    } finally {
      setLoading(false);
    }
  };

  useEffect(() => {
    fetchRepos();
  }, []);

  const filteredRepos = useMemo(() => {
    const query = search.trim().toLowerCase();
    if (!query) return repos;
    return repos.filter((repo) =>
      repo.name.toLowerCase().includes(query)
    );
  }, [repos, search]);

  if (loading) {
    return (
      <main>
        <section>
          <h2>Projects</h2>
          <p>Loading repositories...</p>
        </section>
      </main>
    );
  }

  if (error) {
    return (
      <main>
        <section>
          <h2>Projects</h2>
          <p>{error}</p>
          <button onClick={fetchRepos}>Retry</button>
        </section>
      </main>
    );
  }

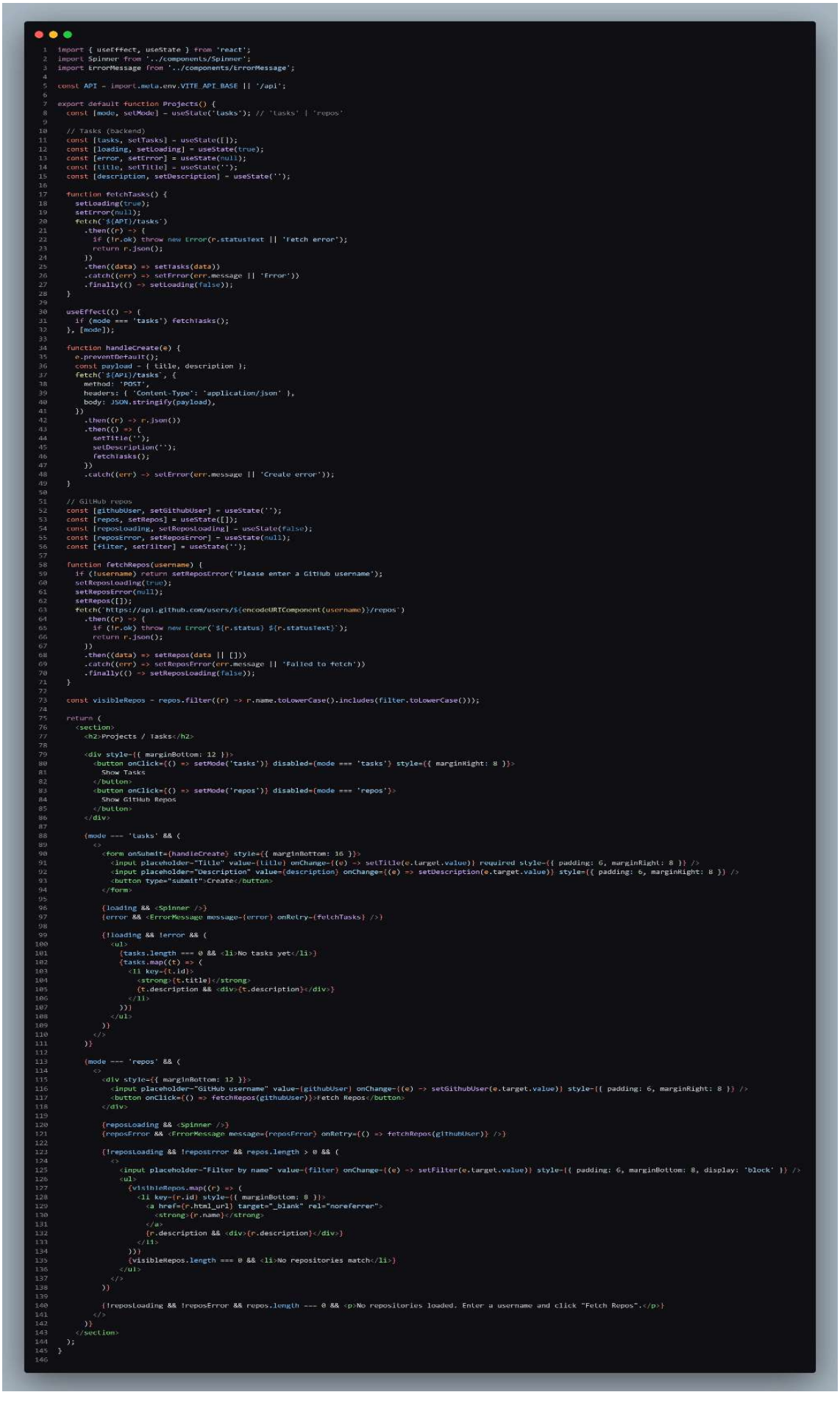
  return (
    <main>
      <section>
        <h2>Projects</h2>

        <input
          type="text"
          placeholder="Search repositories by name..."
          value={search}
          onChange={(e) => setSearch(e.target.value)}
        />

        <p>Total repositories: {filteredRepos.length}</p>

        <ul>
          {filteredRepos.map((repo) => (
            <li key={repo.id}>
              <a href={repo.html_url} target="_blank" rel="noreferrer">
                {repo.name}
              </a>{" "}
              — ⭐ {repo.stargazers_count}
            </li>
          ))}
        </ul>
      </section>
    </main>
  );
}

export default Projects;
```



```jsx
import { useEffect, useState } from 'react';
import Spinner from '../components/Spinner';
import ErrorMessage from '../components/ErrorMessage';

const API = import.meta.env.VITE_API_BASE || '/api';

export default function Projects() {
  const [mode, setMode] = useState('tasks'); // 'tasks' | 'repos'

  // Tasks (backend)
  const [tasks, setTasks] = useState([]);
  const [loading, setLoading] = useState(true);
  const [error, setError] = useState(null);
  const [title, setTitle] = useState('');
  const [description, setDescription] = useState('');

  function fetchTasks() {
    setLoading(true);
    setError(null);
    fetch(`${API}/tasks`)
      .then((r) => {
        if (!r.ok) throw new Error(r.statusText || 'Fetch error');
        return r.json();
      })
      .then((data) => setTasks(data))
      .catch((err) => setError(err.message || 'Error'))
      .finally(() => setLoading(false));
  }

  useEffect(() => {
    if (mode === 'tasks') fetchTasks();
  }, [mode]);

  function handleCreate(e) {
    e.preventDefault();
    const payload = { title, description };
    fetch(`${API}/tasks`, {
      method: 'POST',
      headers: { 'Content-Type': 'application/json' },
      body: JSON.stringify(payload),
    })
      .then((r) => r.json())
      .then(() => {
        setTitle('');
        setDescription('');
        fetchTasks();
      })
      .catch((err) => setError(err.message || 'Create error'));
  }

  // GitHub repos
  const [githubUser, setGithubUser] = useState('');
  const [repos, setRepos] = useState([]);
  const [reposLoading, setReposLoading] = useState(false);
  const [reposError, setReposError] = useState(null);
  const [filter, setFilter] = useState('');

  function fetchRepos(username) {
    if (!username) return setReposError('Please enter a GitHub username');
    setReposLoading(true);
    setReposError(null);
    setRepos([]);
    fetch(`https://api.github.com/users/${encodeURIComponent(username)}/repos`)
      .then((r) => {
        if (!r.ok) throw new Error(`${r.status} ${r.statusText}`);
        return r.json();
      })
      .then((data) => setRepos(data || []))
      .catch((err) => setReposError(err.message || 'Failed to fetch'))
      .finally(() => setReposLoading(false));
  }

  const visibleRepos = repos.filter((r) => r.name.toLowerCase().includes(filter.toLowerCase()));

  return (
    <section>
      <h2>Projects / Tasks</h2>

      <div style={{ marginBottom: 12 }}>
        <button onClick={() => setMode('tasks')} disabled={mode === 'tasks'} style={{ marginRight: 8 }}>
          Show Tasks
        </button>
        <button onClick={() => setMode('repos')} disabled={mode === 'repos'}>
          Show GitHub Repos
        </button>
      </div>

      {mode === 'tasks' && (
        <>
          <form onSubmit={handleCreate} style={{ marginBottom: 16 }}>
            <input placeholder="Title" value={title} onChange={(e) => setTitle(e.target.value)} required style={{ padding: 6, marginRight: 8 }} />
            <input placeholder="Description" value={description} onChange={(e) => setDescription(e.target.value)} style={{ padding: 6, marginRight: 8 }} />
            <button type="submit">Create</button>
          </form>

          {loading && <Spinner />}
          {error && <ErrorMessage message={error} onRetry={fetchTasks} />}

          {!loading && !error && (
            <ul>
              {tasks.length === 0 && <li>No tasks yet</li>}
              {tasks.map((t) => (
                <li key={t.id}>
                  <strong>{t.title}</strong>
                  {t.description && <div>{t.description}</div>}
                </li>
              ))}
            </ul>
          )}
        </>
      )}

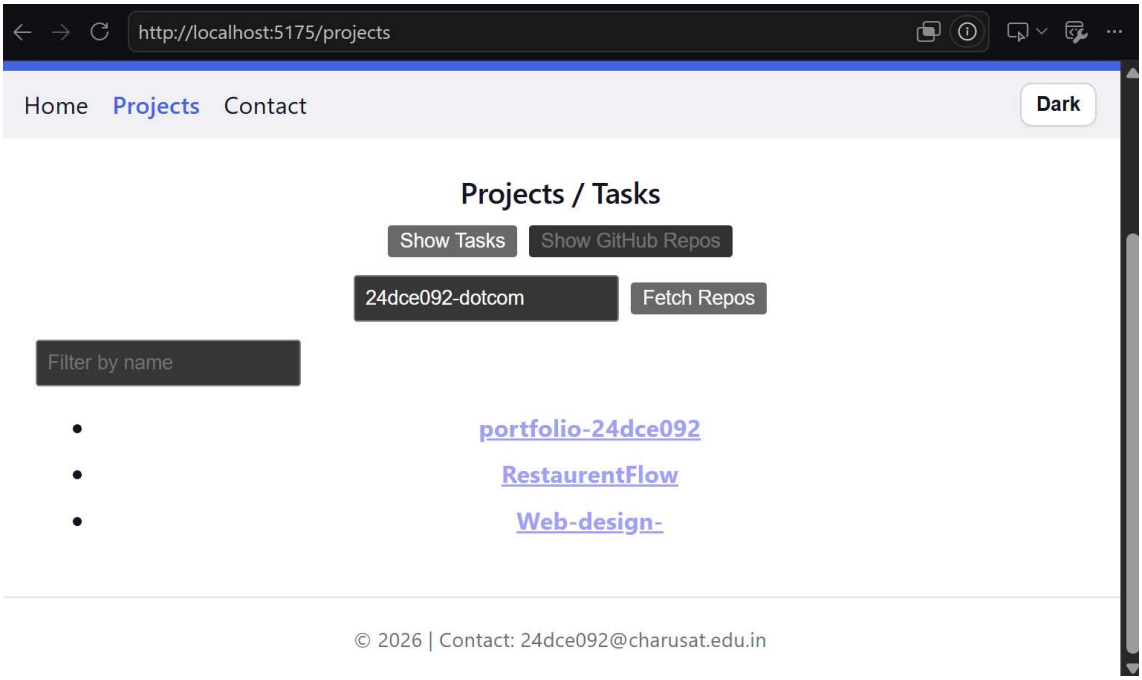
      {mode === 'repos' && (
        <>
          <div style={{ marginBottom: 12 }}>
            <input placeholder="GitHub username" value={githubUser} onChange={(e) => setGithubUser(e.target.value)} style={{ padding: 6, marginRight: 8 }} />
            <button onClick={() => fetchRepos(githubUser)}>Fetch Repos</button>
          </div>

          {reposLoading && <Spinner />}
          {reposError && <ErrorMessage message={reposError} onRetry={() => fetchRepos(githubUser)} />}

          {!reposLoading && !reposError && repos.length > 0 && (
            <>
              <input placeholder="Filter by name" value={filter} onChange={(e) => setFilter(e.target.value)} style={{ padding: 6, marginBottom: 8, display: 'block' }} />
              <ul>
                {visibleRepos.map((r) => (
                  <li key={r.id} style={{ marginBottom: 8 }}>
                    <a href={r.html_url} target="_blank" rel="noreferrer">
                      <strong>{r.name}</strong>
                    </a>
                    {r.description && <div>{r.description}</div>}
                  </li>
                ))}
                {visibleRepos.length === 0 && <li>No repositories match</li>}
              </ul>
            </>
          )}

          {!reposLoading && !reposError && repos.length === 0 && <p>No repositories loaded. Enter a username and click "Fetch Repos".</p>}
        </>
      )}
    </section>
  );
}
```

Output :


http://localhost:5175/projects
Home
Projects
Contact
Dark
Projects / Tasks
Show Tasks
Show GitHub Repos
24dce092-dotcom
Fetch Repos
Filter by name
portfolio-24dce092
RestaurentFlow
Web-design-
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